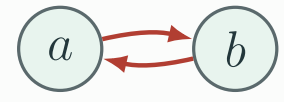


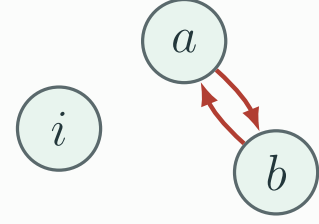
Problem and Contribution

Problem: too weak



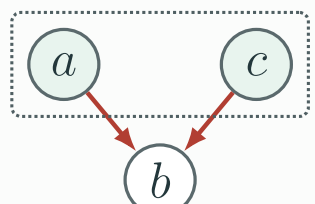
In symmetric AFs, admissible sets collapse to conflict-free sets.

Problem: too restrictive



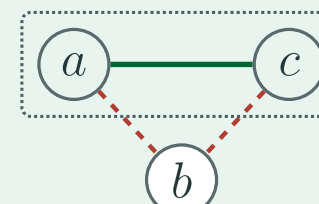
Strong admissibility excludes conflict-involved actors; only isolated points survive.

AF extension



jointly acceptable: conflict-free and defended by the set

Social extension



cohesive community: allies answer conflicts; brokerage reaches external conflicts

Reading. The starting point is a semantic mismatch in symmetric conflict. Dung admissibility is too weak because, in symmetric AFs, admissible sets collapse to conflict-free sets; strong admissibility is too restrictive because it keeps only isolated actors. External Semantics repairs this with **strict third-party defence**. This repaired AF notion is then used as a bridge to undirected networks: an AF extension is a jointly acceptable set, and a social extension is reinterpreted as a cohesive community whose allies can answer conflicts and later broker external conflicts through positive ties.

External Semantics: Definition and Baseline

The negative layer induces a symmetric argumentation framework

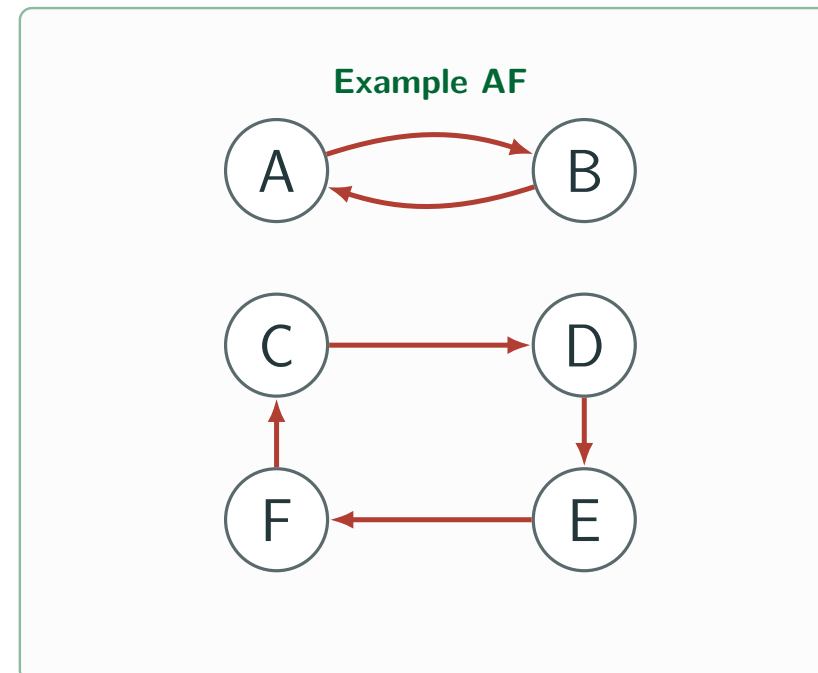
$$AF^-(\mathcal{N}) = \langle A, \rightarrow \rangle, \quad a \rightarrow b \text{ iff } aR^-b.$$

For $AF = \langle A, \rightarrow \rangle$ and $S \subseteq A$, S **externally defends** $a \in A$ iff

$$\forall b (b \rightarrow a \Rightarrow \exists c \in S \setminus \{a\} : c \rightarrow b).$$

External admissibility requires S to be conflict-free and to satisfy this strict third-party defence condition for every selected actor.

Why introduce it?



Dung admissible sets

$\emptyset$; $\{A\}$, $\{B\}$; $\{C, E\}$, $\{D, F\}$; plus $\{A\}$ or $\{B\}$ with either pair.

Strong admissible sets

only $\emptyset$: self-return and circular defence are rejected.

External admissible sets

$\emptyset$, $\{C, E\}$, $\{D, F\}$: defence comes from a distinct third party.

External Semantics keeps a middle route: the baseline requirement is not mere absence of internal conflict, but **strict third-party defence**. A selected actor must be defended by a distinct member of the same group.

From Abstract Defence to Social Brokerage

Symmetric conflict reciprocal attacks

Dung defence becomes self-opposition; strong admissibility becomes too restrictive.

External Semantics strict third-party defence

A selected actor needs a distinct ally answering each attacker.

Social reading acceptable communities

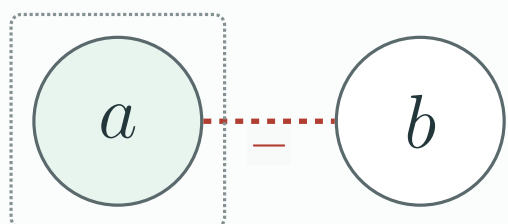
Extensions become groups that can jointly withstand social conflict.

Conflict Brokerage positive access to conflicts

Positive ties make brokerage a specifically social-network role.

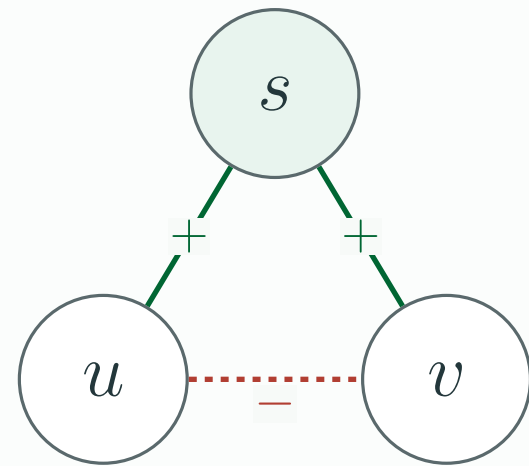
Core Examples: Three Roles and the Brokerage Refinement

(a) Self-defence



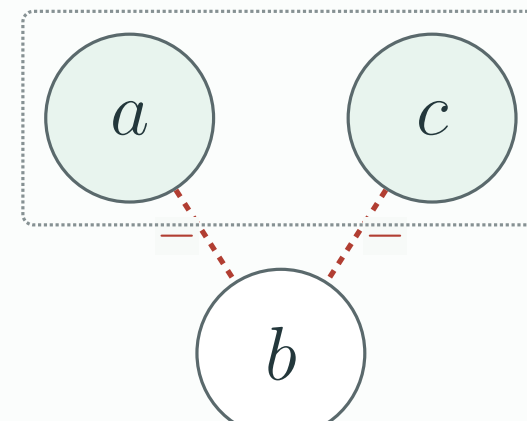
$\{a\}$: reciprocal conflict is not a third-party answer

(b) Brokerage



one witness reaches both endpoints

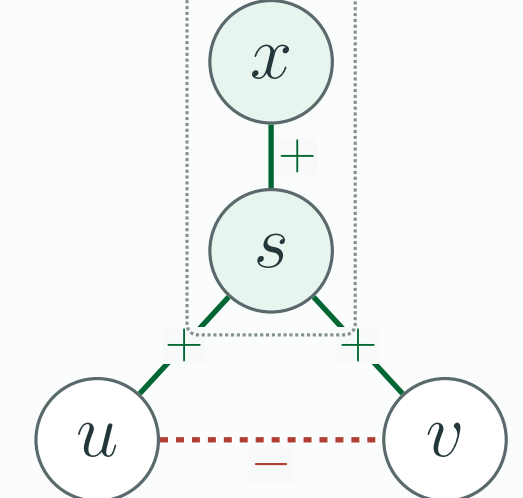
(c) Alliance



$S = \{a, c\}$: external alliance

refine

(d) Free-rider test



same Br_w : remove the free rider

Reading. (a) self-return is not defence; (c) an ally gives the missing third-party answer; (b) brokerage requires positive access to both endpoints; (d) the further refinement removes free riders.

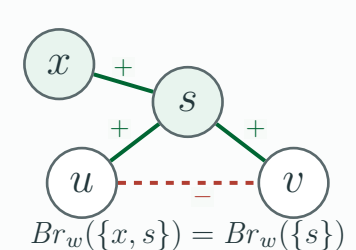
Conflict Brokerage Semantics: Basic and Further Levels

Object. After the external baseline, the paper evaluates a social-network role: which communities can broker conflicts using positive access?

For $S \subseteq A$, a negative edge is external to S when both endpoints lie outside S :

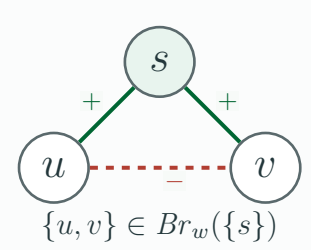
$$Br_w(S) = \{\{u, v\} \mid u, v \in A \setminus S, uR^-v, \exists s \in S : sR^+u \wedge sR^+v\}.$$

free-rider candidate



$Br_w(\{x, s\}) = Br_w(\{s\})$

brokerage core



$\{u, v\} \in Br_w(\{s\})$

1. Candidate

conflict-free + positively connected + $Br_w(S) \neq \emptyset$

2. Basic extension

no candidate has a strictly larger witnessed range

3. Further level

no proper subset preserves feasibility and the same Br_w

Social-network specificity. The new semantics needs an independent positive layer R^+ ; in a plain abstract argumentation framework, there is no social-access relation with which to witness brokerage.

Meaning. Basic brokerage can carry a free rider x that adds no witnessed conflict. The further level keeps the smaller core s , which preserves the same Br_w .

OHA AI Research Agenda: Current Work and Next Stages

I symmetric conflict why admissibility fails

II External Semantics third-party defence

III bipolar brokerage positive social access

IV aggregation + dynamics repair and change

V data-driven models predict extensions

Current work. External Semantics gives the formal baseline for third-party defence; Conflict Brokerage Semantics is the social-network step built on that baseline. The current paper therefore explains why symmetric conflicts need external defence, then uses positive relations to evaluate which communities can mediate external conflicts without carrying free riders.

Relation to related work. The agenda builds on Dung-style acceptability and symmetric frameworks, compares with strong and cooperative defence, and connects social argumentation with brokerage and structural holes. It treats External Semantics as the bridge from abstract argumentation to social networks, while the brokerage refinement marks the specifically social-network part of the programme.

Selected references.

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Yi and Chen, 2026, *External Semantics for Argumentation Frameworks*.

Camina and Dunne, 2020, *Strong admissibility revisited*.
Blümel and Ulbricht, 2026, *Defining defense and defeat from scratch*.
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Leite and Martins, *Social Abstract Argumentation*.
Burt, 1992, *Structural holes*.
Craandijk and Bex, 2020, *Deep learning for abstract argumentation*.